Cosmetics Store Data Analysis: Full Project Documentation

Tools: SQL Server (T-SQL, SSMS), Python, Power BI Data period: November 2025 to September 2026

1. Project Overview

This project is an end-to-end data analysis for a real cosmetics store selling makeup, skincare, haircare, perfumes, accessories and other beauty products. The store's system exports were raw and inconsistent, so the work covers the full path from raw files to validated, analysis-ready data and then to management insights.

The project follows a strict staged workflow: Raw Data, Understanding, Cleaning, Validation, Preparation, and an Insight-Ready dataset. The insight and exploratory analysis phase started only after the earlier stages were reviewed and approved.

Business problem

The store owner needs to understand why sales are changing, where cash is tied up in inventory, which products and categories create revenue and profit, and which data problems could lead to wrong decisions.

Objectives

- Build a clean, validated and documented dataset from six raw exports.
- Explain the change in sales through traffic (invoices per day), basket size and unit price.
- Measure product availability and its relationship with sales.
- Quantify inventory capital tied up in slow, dead or never-sold products.
- Rank products and categories by revenue, gross profit and margin.
- Analyse returns and shopping patterns by hour and weekday.
- Record data quality issues and show how they affect decisions.

Scope

- Retail sales net of returns, for November 2025 to September 2026.
- Gross profit only. Operating expenses are not available, so net profit and break-even are out of scope.

- Customer-level analysis is out of scope because most invoices carry no customer name.

Tools and approach

- **SQL Server (T-SQL, SSMS):** cleaning, validation, transformation and reporting views.
- **Python:** supporting cleaning and checks.
- **Power BI and Tableau:** planned visualization and reporting layer on top of the SQL views.
- All cleaning was done with SQL and Python, not Excel. Raw, cleaned and transformed datasets are stored separately, and raw files were never overwritten.

2. Data Sources

Source file	Content	Role in the analysis
Sales (بيع)	Sales invoice lines	Revenue, traffic, basket size, margin
Purchases (شراء)	Supplier purchase invoices and quantities	Replenishment activity, availability
Returns (مرتجع_بيع)	Sales returns	Net sales, return analysis
Inventory / Products (البضاعه)	Product list and stock balances	Stock value, stock status, availability
Accounts (الحسابات)	Customers, suppliers, internal accounts	Account classification
Profit and Loss (الارباح_والخسائر)	Store profit and loss export	Financial context and reconciliation reference

The Returns file was provided separately. Sales analysis was not finalised until returns were integrated and validated.

3. Data Preparation: Cleaning and Validation

3.1 Workflow

1. **Raw Data:** the six files are stored as received and kept unchanged.
2. **Understanding:** structure, fields, relationships and anomalies are reviewed for each file.

3. **Cleaning:** formatting, naming and duplicate issues are handled in SQL and Python.
4. **Validation:** quantities, prices, invoice sequences, stock balances and returns are checked.
5. **Preparation:** standardised tables and derived fields are built.
6. **Insight-ready layer:** analysis datasets and reporting views are produced.

3.2 Key data decisions

Each decision below was made explicitly to avoid damaging valid business data or inventing facts.

Issue	Decision	Reason
Sales lines with zero price	Kept as valid transactions and flagged	They are genuine gifts or giveaways, not errors
Fractional sales quantities	Kept as valid	They represent partial sets or bundles
Product names that differ only by spacing or formatting	Treated as one product through a mapping table (Original Name, Standard Name, Product Code) built before merging	Prevents split product history. Uncertain cases are not merged
No reliable product category in the source	Added an `inferred_category` column derived from product names (Makeup, Skincare, Haircare, Perfumes, Accessories, Nails, Tools, Personal Care, Gift/Packaging, Other/Unknown)	The original category column is kept unchanged, the inferred values are documented as analyst-inferred, and ambiguous names stay as Other/Unknown
Internal accounts (partner, staff and payroll, store expense accounts)	Kept in the data with a separate classification and excluded from customer and supplier analysis	Prevents internal movements from inflating customer metrics. The classification logic is documented
Missing invoice numbers in the sequence	Logged as "Missing Invoice Sequence - Status Unknown" in a separate data quality log	The cause is unknown, so they are not assumed to be cancellations, not fabricated, and existing invoices are not altered

Issue	Decision	Reason
Returns	Integrated from the separate returns file before sales analysis was finalised	Sales must be measured net of returns

3.3 Validation checks and data quality findings

- **Returns integration:** net retail sales are calculated after returns.
- **Invoice sequence:** gaps are logged rather than corrected.
- **Stock balances:** 84 products have negative stock, which points to recording errors or sales entered before purchases. A physical count is recommended.
- **Customer identity:** about 98.4% of invoices carry no named customer, and the sales representative field is empty. Retention and sales-rep performance therefore cannot be measured.
- **Category quality:** categories are inferred, and the share of Other/Unknown should be reported next to category results.
- **Pending validation:** a documented reconciliation of sales and returns against the Profit and Loss export is still to be added.

4. Results and Insights

The results below come from validated data for November 2025 to September 2026. They describe associations in the data and do not prove causes.

4.1 Headline results

Metric	Result
Retail net sales (after returns)	901,987 EGP
Net gross profit	192,534 EGP (before operating expenses)
Net gross margin	21.3%
Invoices per trading day	29.3 to 10.2 (December 2025 to August 2026, about 65% lower)
Average unit price	72.9 to 73.7 EGP, broadly flat
Catalogue availability	97.9% to 62.9% (November 2025 to September 2026)

Metric	Result
Inventory at cost	About 336,000 EGP

4.2 Key insights

1. **Traffic fell much more than price changed.** Invoices per day dropped about 65% while the average unit price stayed almost flat. This points to traffic rather than pricing as the main change. December is the launch period, so part of the drop may be a return to normal levels.
2. **Availability weakened as purchasing slowed.** Units purchased fell from 2,865 in December to 25 in June and 3 in August, while catalogue availability fell from 97.9% to 62.9%. The two move together, which is consistent with lost sales from unavailable products, but the data cannot prove the direction of cause.
3. **Restocking is associated with higher sales.** After three large restocks, revenue of the replenished products changed by +108%, +78% and +129%, against +11%, -4% and -34% for other products. With only three cases, and restocked items being proven sellers, this is indicative and not conclusive.
4. **Inventory capital is poorly allocated.** 121 products are overstocked, 153 have never sold and 271 had no sale in the last 90 days, together worth about 250,000 EGP at cost. These groups may overlap, so the combined value should be confirmed as a union.
5. **Stock-outs and slow stock exist together.** 457 products are out of stock while cash is tied up in slow-moving stock.
6. **Margin differs by category (inferred).** Makeup contributes 22.6% of revenue at a 25.6% margin, Personal Care 22.4% of revenue at a 17.5% margin, and Accessories earn a 28.0% margin.
7. **Demand is concentrated in the evening.** About 68% of invoices fall between 19:00 and 01:00, and Friday performs clearly better than Monday.

4.3 Recommendations

Timing	Action
Immediately	Verify and replenish high-margin A products that are out of stock or have low cover, without broad range expansion
Immediately	Pause purchasing of never-sold and no-sale-in-90-day products until each is reviewed

Timing	Action
Within 30 days	Review high-revenue, low-margin products for supplier cost, price and discount leakage
Within 30 days	Start a weekly review of traffic, availability, stock-outs, margin and slow inventory
Next process change	Record customer ID (with consent), sales representative, purchase receipt dates and reasons for stock adjustments
Next financial close	Add rent, payroll, utilities, fees and marketing cost to calculate net profit and break-even

5. Limitations

- The period covers about 11 months, which is too short to establish seasonality.
- Categories are inferred from product names and are not official.
- Only gross profit is available. There are no operating expenses.
- Customer-level analysis is not possible because of missing customer identity.
- Stock balances are not verified by a physical count.
- Findings on traffic, availability and restocking show association, not proven causation.

6. Conclusion and Next Steps

The project turned six raw exports into a validated, documented dataset and a set of management insights. The main message is that sales weakness is linked to falling traffic and shrinking product availability, while a large amount of cash is tied up in slow and dead stock.

Planned next steps are to document the reconciliation with the Profit and Loss export, verify stock with a physical count, and build the dashboard in Power BI and Tableau on top of the SQL views. A simple chatbot will be embedded in the dashboard to explain what each KPI and chart means for the store owner. It explains the dashboard itself and does not answer live questions about the data. Its engine (rule-based or AI-powered) is still to be decided.